

Weighted-norm bounds for the canonical coefficients

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Abstract

Astra #10 rank 4 (R6 boundedness). The weighted coefficient norm at exponent α , log degree ℓ and radius r is $\|c\|_{\alpha,\ell,r} = \sum_{ij} r^{i+j} \int_0^\infty s^{\alpha-1} (1 + |\log s|)^\ell e^{-\beta s^2} |c_{ij}(s)| ds$. For $b < r < \rho$ and c in the envelope class the norm is finite, and the canonical coefficients are bounded by it: $|A_\alpha(c)| \leq w_C(\alpha) \|c\|_{\alpha,0,r}$ and $|B_\alpha(c)| \leq (w_U(\alpha) + w_V(\alpha)) \|c\|_{\alpha,0,r} + w_C(\alpha) \|c\|_{\alpha,1,r}$ (`abs_canonA_le_coeffNorm`, `abs_canonB_le_coeffNorm`) with weights depending only on (α, b, r, h, k) . The key input is the geometric bound $|\text{axisPrim}(\gamma, b, j)| \leq C_\gamma b^j$ on the finite-part multipliers (`abs_axisPrim_le_geom`), exactly as Astra #10 prescribed. Zero sorry/axiom.

1 The things formalised

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noncomputable def coeffNorm ( : ) ( : ) (r : ) (c : × → → ) : :=
  ' ij : × , r ^ (ij.1 + ij.2) * tailMoment      (c ij)
theorem coeffNorm_summable ... (hr0 : 0 < r) (hr : r < ) ... :
  Summable fun ij : × => r ^ (ij.1 + ij.2) * tailMoment      (c ij)
noncomputable def axisPrimGeomConst ( b : ) : :=
  b ^ (-) / ((canonicalM : ) - )
  + m Finset.range (canonicalM ), |axisPrim b m| / b ^ m
theorem abs_axisPrim_le_geom ( b : ) (hb : 0 < b) (j : ) :
  |axisPrim b j| axisPrimGeomConst b * b ^ j
noncomputable def cWeight (r : ) (h h k k : ) ( : ) : :=
  if collision then 1 / ((k : ) * k * r ^ (uIdx h k + vIdx h k )) else 0
noncomputable def uWeight (b r : ) (h h k k : ) ( : ) : :=
  if u-pole then axisPrimGeomConst ((k : ) * - h - 1) b / ((k : ) * r ^ uIdx h k )
  else 0
theorem abs_canonA_le_coeffNorm ... ( : ) (h : 0 < ) :
  |canonA h h k k (fun i j s => c (i, j) s) |
  cWeight r h h k k * coeffNorm 0 r c
theorem abs_canonB_le_coeffNorm ... (hbr : b < r) (hr : r < ) ... ( : ) (h : 0 < ) :
  |canonB b h h k k (anaFaceU c b) (anaFaceV c b) (fun i j s => c (i, j) s) |
  (uWeight b r h h k k + vWeight b r h h k k ) * coeffNorm 0 r c
  + cWeight r h h k k * coeffNorm 1 r c

```

2 Mathematics

Each tail moment is at most $\rho^{-i-j} C_0 EM$ by the envelope, so the weighted sum converges geometrically for $r < \rho$. The collision coefficient is a single scaled entry, bounded by one term of the norm. For the face coefficient at $\alpha = \alpha_i$, the moment series $k_1^{-1} \sum_j q_{\gamma_i}(j) M[\alpha; \ell; c_{ij}]$ (unit 126) is bounded termwise by $C_{\gamma_i} b^j \cdot \text{tailMoment}(c_{ij}) \leq C_{\gamma_i} r^{-i} \cdot r^{i+j} \text{tailMoment}(c_{ij})$, and the row sum is at

most the full weighted sum (comparison along the injection $j \mapsto (i, j)$). The multiplier bound uses $|q_\gamma(j)| \leq b^j b^{-\gamma} / (J - \gamma)$ for $j \geq J > \gamma$ and a finite sum for $j < J$.

3 Paper-facing meaning

The canonical coefficients of the $d = 2$ Taylor tree are bounded functionals of the amplitude coefficient array in a weighted ℓ^1 -of-Gaussian-moments norm, with explicit operator constants. With linearity (next unit) this gives the difference form $|A_\alpha(c) - A_\alpha(d)| \leq w \|c - d\|$, the deterministic half of the continuity needed to pass from $\xi_n \Rightarrow G$ to convergence of the coefficients (Astra #10: R6 is the bridge to convergence in law; the remaining nonlinear link is the amplitude map $(x, y) \mapsto c$).

4 Lean notes

- `Summable.prod_factor` (rows) and `.prod_symm.prod_factor` (columns) extract one-index summability from a summable family on $\mathbb{N}^2$; `Summable.tsum_le_tsum_of_inj` compares a row sum with the full sum.
- `gcongr` descends INTO a `tsum` and leaves termwise goals with inaccessible binder names; when the intended step is a `tsum`-level inequality, use `mul_le_mul_of_nonneg_left` and `Summable.tsum_le_tsum` explicitly.
- `Integrable.div_const` returns `Integrable ... (volume.restrict s)`, on which dot-notation `.congr_fun` (an `IntegrableOn` lemma) fails; restate the type as `IntegrableOn` in a `have` first.
- `Summable.mul_left _` with an underscore can time out at `isDefEq`; pass the constant explicitly.